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May 1, 2026

Editor-in-Chief
IEEE Signal Processing Letters

Dear Editor,

I am resubmitting the manuscript entitled

"When Does Temporal Memory Help? Selective SSM vs. MLP in RL-Based Multi-Target DOA Track Management"

for consideration for publication in *IEEE Signal Processing Letters* as an original Letter.

Resubmission disclosure. This manuscript was previously assigned ID **SPL-46772-2026** and was *withdrawn by the author on 24 April 2026, prior to assignment of any reviewers and before any review reports were generated.* No reviewer feedback was received. The withdrawal was made solely to correct three clerical errors in the reference list that I identified after submission. A separate supporting document (`manuscript_history.pdf`) accompanying this resubmission provides a full disclosure and confirms compliance with IEEE Signal Processing Society Policies and Procedures Manual Section 6.16, whose "one resubmission" limit applies to manuscripts *rejected after review* and is therefore not invoked by this pre-review withdrawal.

What changed since SPL-46772-2026. Three reference entries were corrected against arXiv and IEEE Xplore:

- **Ref [8]** (arXiv:2205.11104) – author list corrected from "N. Dorka" to the actual authors "M. Pleines, M. Pallasch, F. Zimmer, and M. Preuss."
- **Ref [11]** (Ota, Decision Mamba) – title and arXiv ID corrected from "…Hybrid Selective Sequence Modeling, arXiv:2406.00079" to the actual Ota paper "Decision Mamba: Reinforcement Learning via Sequence Modeling with Selective State Spaces, arXiv:2403.19925" (arXiv:2406.00079 is a separate paper by Huang et al. previously conflated with Ota's work).
- **Ref [12]** (Erol et al., Audio Mamba) – title corrected to the published IEEE SPL title "*Bidirectional* State Space Model for Audio Representation Learning" (IEEE Xplore record 10720871).

The technical content – Proposition 1, Algorithm 1, all theoretical derivations, all experimental results, all figures, and all tables – is **unchanged** from the withdrawn version. The corrections are purely bibliographic. All thirteen remaining references were re-verified and confirmed correct.

Summary of contribution. This Letter addresses a fundamental question in reinforcement-learning (RL)-based adaptive multi-target DOA tracking: when does adding temporal memory to an RL policy improve performance over a memoryless multilayer perceptron? Using the data-processing inequality and the notion of a sufficient statistic, we prove that a selective state-space model (SSM) encoder outperforms an MLP if and only if the instantaneous observation is not a sufficient statistic for the policy-relevant history (Proposition 1). We instantiate this result as **Mamba-COP-RL**, coupling a Mamba SSM encoder with a PPO actor–critic operating on raw 181-dimensional COP spatial spectra. On **CombatTrackEnv**, an open-source benchmark with four combat-realistic multi-target scenarios (Jamming, Stealth, Saturation, Formation), Mamba-COP-RL improves GOSPA by up to 25.2% over fixed GM-PHD thresholds and 6.7% over MLP-PPO, with gain magnitudes ordered by temporal-dependency horizon exactly as predicted by the theory. Ablations against GRU-PPO and LSTM-PPO confirm the role of Mamba's input-dependent selectivity. The 41.4 KB INT8 model runs at 3.4 ms/scan on an STM32H7 Cortex-M7, providing a concrete edge-deployable artifact.

Scope. The submission is squarely within the scope of *IEEE Signal Processing Letters*: (i) sensor-array signal processing (higher-order-cumulant DOA estimation), (ii) statistical signal processing for multi-target tracking (random-finite-set / GM-PHD filtering), and (iii) machine learning for signal processing (selective SSMs, RL on signal-domain observations). The principal contribution is a single, sharply stated theoretical condition with a tightly scoped empirical validation – a natural fit for the Letter format.

Related submission cited as Ref [3]. A companion manuscript "*Underdetermined High-Resolution DOA Estimation and Multi-Target Tracking via COP-RFS*" (J. H. Choi, 2026) is currently under review at *IEEE Transactions on Signal Processing* and is cited as Ref [3] in this Letter. That work introduces the COP-RFS signal-processing pipeline; the present Letter adds an RL-based GM-PHD parameter scheduler on top of that pipeline and formally characterizes when temporal memory is beneficial. The two contributions are complementary and non-overlapping in their results, figures, and tables.

Code availability. A reproducible release accompanies this submission. Reviewers can reproduce the headline Table II numbers in approximately 30 seconds on a CPU laptop via a single command, `python reproduce_spl.py`. The repository (including the manuscript PDF placed alongside the reproduction script) is publicly available at `https://github.com/DrJinHoChoi/IronDome-DOA-Tracking` under a dual academic / commercial license. The exact submission commit is git-tagged `spl2026-v1`.

Author contributions and conflicts of interest. This is a sole-authored manuscript; the author conceived the proposition, designed the architecture, implemented the CombatTrackEnv benchmark and Mamba-COP-RL, ran all experiments, and wrote the manuscript. There are no conflicts of interest. No funding was received that requires disclosure.

Declarations. Apart from the previously withdrawn SPL-46772-2026 documented above, the manuscript has not been published previously, is not under review at any other journal or conference, and is not derived from any prior conference publication. No formal preprint has been posted to arXiv, TechRxiv, or any dedicated preprint server; a copy of the PDF is included in the public code repository above for reproducibility (IEEE-permissible self-hosting alongside the reproduction code).

Thank you for considering this resubmission. I am happy to provide any additional clarification the editorial board may require.

Sincerely,

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